
MULTI-CONFIGURATIONAL TIME-DEPENDENT HARTREE-FOCK

Stian Bilek
Department of Physics
University of Oslo
Oslo, Norway
`stian.bilek@fys.uio.no`

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ABSTRACT

In this project, we explored the dynamics of two electrons in a quantum harmonic dot using the spin-restricted Multi-Configurational Time-Dependent Hartree-Fock (MCTDHF) method. Our findings were compared to those of Zanghellini et al. [1], who used a spin-unrestricted version. Despite the differences in spin handling, our results closely aligned with theirs under specific conditions: Using one MCTDHF orbital per spin-subspace (similar to Hartree-Fock) and when nearing the optimal ground state energy determined by the spin-restricted configuration interaction method. This correspondence was evident in particle densities and time evolution patterns. However, exactly replicating their results proved challenging due to our different approach to spin.

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1 Introduction

A complete description of any quantum system beyond the simplest requires the use of sophisticated computational techniques. Although many such techniques have been developed, one often faces a trade-off between the complexity of the algorithm and the accuracy of the desired approximation. For example, the Hartree-Fock algorithm (see [2] chapter 3), which attempts to describe the ground state using only one Slater determinant, is computationally efficient but falls short for highly correlated systems. Another example is the full configuration interaction algorithm (see [2] chapter 4), which provides a comprehensive description of the system within a specified basis but scales poorly as the degrees of freedom of the system increase. One reason for the excitement surrounding quantum computers is the hope that they will enable us to simulate quantum systems both accurately and efficiently, as emphasized by Richard Feynman's famous assertion: "...nature isn't classical, dammit, and if you want to simulate nature, you better make it quantum mechanical..."[3]. However, the technology has not yet reached the stage where it can outperform classical computers in practical tasks. Additionally, and perhaps ironically, the engineering of fault-tolerant quantum devices may require classical simulation of time-dependent quantum systems to reduce the error rates of one- and two-qubit gates. Should quantum computers fail to fulfill their promise, we remain dependent on efficient and accurate classical algorithms to further our understanding of quantum mechanical systems.

The Multi-Configurational Time-Dependent Hartree-Fock (MCTDHF) method is a promising advancement in the field of quantum mechanics simulations on classical computers, particularly for time-dependent systems. Unlike the traditional Hartree-Fock approach, which uses a fixed basis set to describe electron configurations, MCTDHF employs a time-dependent basis set. This basis set evolves in response to external perturbations, allowing for a more accurate representation of dynamics and interactions. The adaptability of the basis set could be particularly beneficial for capturing the behavior of systems subjected to time-varying external fields.

In this project, we aim to explore the theoretical aspects of the MCTDHF method, its computational implementation, and its application to problems in quantum physics. We will implement the spin-restricted MCTDHF method for quantum systems in one dimension and benchmark our results against the Configuration Interaction (CI) method and the results of Zanghellini *et al* [1] for the harmonic quantum dot. The latter problem consists of finding the ground state energy of two electrons in a Harmonic oscillator potential and then expose these electrons to a time-dependent monochromatic laser field.

The project is structured as follows: In the theory section 2 we will start by explaining the main ingredients of the MCTDHF method, starting with the Schrödinger equation (2.1), single particle basis functions (2.2) and Slater determinants (2.3). The full configuration interaction (FCI) method and second quantization formalism is then explained in sections 2.4 and 2.5, respectively, before finally explaining the MCTDHF method in section 2.6. In the method section 3 we will go through the mathematical formulation of our one- and two-body potential and laser field in subsection 3.1. We will then detail the construction of our single particle basis in subsection 3.2, and discuss the time evolution and numerical integration in subsection 3.3. Our results will be presented in the results and discussion section 4, before finally concluding in section 5.

The code utilized to produce the results can be found at <https://github.com/stiandb/MCTDHF>.

2 Theory

In the following subsections, we will present the theory required to implement the MCTDHF method. We will first start constructing our basic tool box by explaining the utilization of single particle basis sets and Slater determinants as the ingredients of many-body quantum states and describing the second quantization formalism. The first subsections are highly inspired by the book *Modern Quantum Chemistry: Introduction to Advanced Electronic Structure Theory* by Attila Szabo and Neil S. Ostlund [2]. The derivation of the MCTDHF method is located in section 2.6 and closely follows the work by Hochstuhl *et al* [4].

2.1 The time independent Schrödinger equation

Despite our intention of solving the time-dependent Schrödinger equation, the method we will eventually describe also relies on dealing with the time-independent form of the equation. That is,

$$\hat{H} |\psi_i\rangle = E_i |\psi_i\rangle,$$

where $\hat{H}$ is the Hamiltonian of our system, $|\psi_i\rangle$ is its i 'th eigenstate and E_i is the corresponding energy. The stationary Hamiltonian we will consider consists of one or more electrons trapped in an external potential. Each electron thus posses kinetic energy, they interact with each other via the Coloumb potential while also interacting with the external potential. The corresponding Hamiltonian for N electrons can be written in atomic units as

$$\begin{aligned} \hat{H} &= - \sum_{i=1}^N \frac{1}{2} \hat{\nabla}_i^2 + \sum_i V(\hat{r}_i) + \sum_{i=1}^N \sum_{j>i}^N \frac{1}{\hat{r}_{ij}} \\ &= \sum_i \hat{h}_i + \sum_{i=1}^N \sum_{j>i}^N \hat{v}_{ij} \end{aligned} \quad (1)$$

where $\hat{r}_i$ is the position of the i 'th electron and $\hat{r}_{ij}$ is the distance between electron i and j . In the first format, the first term describes the kinetic energy of each electron, the second term describes the effect of the external potential on each electron, while the third term describes the Coloumb interaction between electron i and j . In the second format, $\hat{h}_i$ is the kinetic and potential energy of electron i , while $\hat{v}_{ij}$ is the Coloumb interaction between electron i and j .

Unfortunately, there is no known analytical solution to this problem when dealing with an arbitrary number of electrons. The numerical method we will apply instead requires us to go through some formalisms, which is what we will do in the following sections.

2.2 Single particle basis

The concern of this section is to describe the construction of the many-electron wave function of our system. To do this, we will exploit the fact that an arbitrary function $g(\mathbf{r})$ could be exactly expanded as a linear combination of an orthonormal and complete set of functions $\langle \psi_i | \psi_j \rangle = \int d\mathbf{r} \psi_i^*(\mathbf{r}) \psi_j(\mathbf{r}) = \delta_{ij}$. That is,

$$g(\mathbf{r}) = \sum_{i=1}^{\infty} c_i \psi_i(\mathbf{r}),$$

where c_i are constants. As the sum indicates, we generally need an infinite number of functions for the set to be complete. However, this is not computationally doable, which means that we will have to make due with a finite set and "hope" that the corresponding approximate solution is close enough. In the context of solving the Schrödinger equation for one particle with position $\mathbf{r}$, the function $g(\mathbf{r})$ can be used as the ansatz for the eigenstates of the Hamiltonian, while the coefficients c_i are taylored to obtain the best possible solution. In this scenario, the functions of the orthonormal set $\{\psi_i\}$ are referred to as spatial orbitals, as they describe the spatial behaviour of our wave-function. If the particle is an electron we will also have to specify its spin. This is done by introducing two orthonormal functions $\alpha(\omega)$ and $\beta(\omega)$, corresponding to spin up and spin down, respectively. Accordingly, we are able to construct our one-electron basis states, called spin orbitals,

$$|\chi(\mathbf{x})\rangle = \begin{cases} |\psi(\mathbf{r})\rangle |\alpha(\omega)\rangle \\ \text{or} \\ |\psi(\mathbf{r})\rangle |\beta(\omega)\rangle \end{cases}.$$

Not unlike the spatial orbitals, spin orbitals also form an orthonormal set $\langle \chi_i | \chi_j \rangle = \int d\mathbf{x} \chi_i^*(\mathbf{x}) \chi_j(\mathbf{x}) = \delta_{ij}$ due to the orthonormality of the spin and spatial components.

2.3 Slater Determinants

While the previous subsection only described how we can generate an arbitrary one-electron wave function, we can fortunately utilize the spin orbitals to construct a basis for a system of N electrons. These N -electron wave functions will have to obey the antisymmetry principle,

$$\Psi(\mathbf{x}_1, \dots, \mathbf{x}_p, \dots, \mathbf{x}_q, \dots, \mathbf{x}_N) = -\Psi(\mathbf{x}_1, \dots, \mathbf{x}_q, \dots, \mathbf{x}_p, \dots, \mathbf{x}_N),$$

stating that the positional exchange of any two electrons changes the sign of the wave function. A way of constructing such a wave function is through what is commonly referred to as Slater determinants, that is

$$\Psi(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \chi_1(\mathbf{x}_1) & \chi_2(\mathbf{x}_1) & \cdots & \chi_N(\mathbf{x}_1) \\ \chi_1(\mathbf{x}_2) & \chi_2(\mathbf{x}_2) & \cdots & \chi_N(\mathbf{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \chi_1(\mathbf{x}_N) & \chi_2(\mathbf{x}_N) & \cdots & \chi_N(\mathbf{x}_N) \end{vmatrix},$$

where $1/\sqrt{N!}$ is a normalization factor. In the Slater determinant, the N electrons are occupying N spin orbitals, but as electrons are indistinguishable, there is no specification as to which electron is in which spin orbital. The antisymmetry is obtained since exchanging two arbitrary columns in a determinant equates to changing its sign. To simplify further discussions, we will use the following shorthand notations for a Slater determinant,

$$|\chi_1(\mathbf{x}_1)\chi_2(\mathbf{x}_2)\cdots\chi_N(\mathbf{x}_N)\rangle = |\chi_1\chi_2\cdots\chi_N\rangle = |ij\cdots k\rangle,$$

hence only highlighting its diagonal elements. Given $2K$ spin orbitals and N electrons, we are able to construct $\binom{2K}{N}$ Slater determinants. Fortunately, this basis is orthonormal as long as the spin orbitals are. That is, given two Slater determinants, $|K\rangle$ and $|J\rangle$, we have that

$$\langle K|J\rangle = \langle ij\cdots k|i'j'\cdots k'\rangle = \delta_{ii'}\cdots\delta_{kk'} = \delta_{KJ}.$$

We have assumed optimal matching of the columns in the two Slater determinants. See Appendix A for a proof of the orthonormality. Being orthonormal, the $\binom{2K}{N}$ Slater determinants can be used as a truncated basis from which we can construct our N -particle wave function.

2.4 Full configuration interaction

Even though the N -particle basis is not complete with a finite number of spin orbitals, we can apply a method called full configuration interaction (FCI) to obtain results that are exact within the basis spanned by our Slater determinants. To do this, we will have to rewrite our Hamiltonian in terms of our N -particle basis and then calculate its eigenvalues and eigenstates. We first write our Hamiltonian (Eq. 1) in terms of a sum of one-electron operators,

$$\hat{O}_1 = \sum_{i=1}^N \hat{h}(\mathbf{r}_i), \quad (2)$$

and a sum of two-electron operators

$$\hat{O}_2 = \sum_{i=1}^N \sum_{j>i}^N \hat{v}(\mathbf{r}_i, \mathbf{r}_j). \quad (3)$$

We are interested in evaluating the matrix elements $\langle K|\hat{O}|J\rangle$ for $|K\rangle = |ij\cdots k\rangle$ and $|J\rangle = |i'j'\cdots k'\rangle$, and we will consider three different scenarios for the states $|K\rangle$ and $|J\rangle$. The first scenario is when $|K\rangle = |J\rangle$ requiring the evaluation of $\langle K|\hat{O}|K\rangle$ for $|K\rangle = |\cdots ij\cdots\rangle$. In the second scenario, we evaluate $\langle K|\hat{O}|J\rangle$ when the Slater determinants are differing by only one spin orbital, that is $|K\rangle = |\cdots ij\cdots\rangle$ and $|J\rangle = |\cdots ik\cdots\rangle$. In the final scenario, we consider the corresponding evaluation when the Slater determinants are differing by just two spin orbitals, that is $|K\rangle = |\cdots ij\cdots\rangle$ and $|J\rangle = |\cdots kl\cdots\rangle$. Fortunately, it turns out that the matrix elements become zero for all other cases. See appendix B for details and a derivation of the matrix elements. The results for the one and two-electron operators can be seen in Table 1.

Having obtained these matrix elements, we are able to calculate $\langle ij\cdots|\hat{H}|kl\cdots\rangle = \langle ij\cdots|\hat{O}_1 + \hat{O}_2|kl\cdots\rangle$ and thus find the eigenstates and eigenvalues of the resulting matrix. The eigenstates $\hat{H}|\psi^{(j)}\rangle = E^{(j)}|\psi^{(j)}\rangle$ will be a linear combination of our Slater determinants,

$$|\psi^{(j)}\rangle = \sum_I C_I^{(j)} |I\rangle,$$

Table 1: Table showing how to calculate the matrix elements for the one-body and two-body operator in Eqs. 2 and 3. We use the notation $\langle i | \hat{h} | j \rangle = \int d\mathbf{x}_1 \chi_i^*(\mathbf{x}_1) \hat{h}(\mathbf{r}_1) \chi_j(\mathbf{x}_2)$ for the one-electron integrals while we use $\langle ij | kl \rangle = \int d\mathbf{x}_1 d\mathbf{x}_2 \chi_i^*(\mathbf{x}_1) \chi_j^*(\mathbf{x}_2) \hat{v}(\mathbf{r}_1, \mathbf{r}_2) \chi_k(\mathbf{x}_1) \chi_l(\mathbf{x}_2)$ for the two-electron integrals.

	$ K\rangle$	$ J\rangle$	$\langle K \hat{O}_1 J \rangle$	$\langle K \hat{O}_2 J \rangle$
Scenario 1:	$ ij \dots\rangle$	$ ij \dots\rangle$	$\sum_p^N \langle p \hat{h} p \rangle$	$\frac{1}{2} \sum_p^N \sum_q^N [\langle pq pq \rangle - \langle pq qp \rangle]$
Scenario 2:	$ ij \dots\rangle$	$ kj \dots\rangle$	$\langle i \hat{h} k \rangle$	$\sum_j^N [\langle ij kj \rangle - \langle ij jk \rangle]$
Scenario 3:	$ ij \dots\rangle$	$ kl \dots\rangle$	0	$\langle ij kl \rangle - \langle ij lk \rangle$

where the sum over I runs through all the Slater determinants in our basis.

We have explained the usage of Slater determinants to construct N -electron wave functions and their potential in finding the eigenvalues and eigenstates of our Hamiltonian. However, as we confront systems with varying particle numbers and interactions, the need for a more elegant formalism is needed. This leads us to Second Quantization, which will be covered in the next section.

2.5 Second Quantization

Second quantization introduces operators designed to encapsulate the inherent antisymmetry principle within fermionic systems. These operators, termed creation and annihilation operators, are linked to the spin orbitals within a Slater determinant. The creation operator $a_i^\dagger$ can be associated with spin orbital χ_i . Its action on a Slater determinant is the following

$$a_i^\dagger |\chi_j \chi_k \dots\rangle = \begin{cases} |\chi_i \chi_j \chi_k \dots\rangle & \text{if } \chi_i \notin |\chi_j \chi_k \dots\rangle \\ 0 & \text{if } \chi_i \in |\chi_j \chi_k \dots\rangle. \end{cases}$$

That is, it creates an electron in orbital χ_i if the orbital is not occupied in the Slater determinant.

We also have its complex conjugate, the annihilation operator a_i , which also can be associated with spin orbital χ_i . Its action on a Slater determinant is to remove an electron from orbital χ_i , if present. That is

$$\begin{aligned} a_i |\chi_i \chi_j \chi_k \dots\rangle &= |\chi_j \chi_k \dots\rangle, \\ a_i |\chi_j \chi_k \dots\rangle &= 0. \end{aligned}$$

When applying the annihilation operator, we assume that the spin orbital associated with the operator is in the first diagonal entry of the Slater determinant. If not, we exchange the columns and introduce the appropriate sign.

The commutation relations between these creation and annihilation operators capture all the properties of Slater determinants. While we simply list these relations without justification here, a detailed derivation can be found in appendix C. The relation will be written in terms of the anticommutator of two operators, $\hat{A}$ and $\hat{B}$, defined by

$$\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}.$$

We first have the commutation relation between two creation operators

$$\{a_i^\dagger, a_j^\dagger\} = a_i^\dagger a_j^\dagger + a_j^\dagger a_i^\dagger = 0. \quad (4)$$

Since this relation states that $a_i^\dagger a_j^\dagger = -a_j^\dagger a_i^\dagger$, we are able to interchange two creation operators as long as we introduce a sign. We also see that if $i = j$, we get $a_i^\dagger a_i^\dagger = -a_i^\dagger a_i^\dagger = 0$, which means that we cannot create two electrons in the same orbital.

The next relation is between two annihilation operators. That is

$$\{a_j, a_i\} = a_j a_i + a_i a_j = 0. \quad (5)$$

As was the case with the creation operators, the annihilation operators are also interchangeable with the introduction of a sign, that is, $a_j a_i = -a_i a_j$. When $i = j$, we get the result that we cannot remove the same electron from a spin orbital twice.

The final relation is between one creation and one annihilation operator. It is given by

$$\{a_i, a_j^\dagger\} = a_i a_j^\dagger + a_j^\dagger a_i = \delta_{ij}. \quad (6)$$

Since $a_i a_j^\dagger = -a_j^\dagger a_i$ for $i \neq j$, we see that we are able to interchange the creation and annihilation operator by the introduction of a sign, as long as they are associated with two different spin orbitals. If $i = j$, we can use the relation $a_i a_i^\dagger = 1 - a_i^\dagger a_i$.

2.5.1 Second-Quantized Operators

Second quantization allow us to represent the one-electron operator in Eq. 2 and the two-electron operator in Eq. 3 in terms of the annihilation and creation operators. That is

$$\hat{O}_1 = \sum_{ij} \langle i | \hat{h} | j \rangle a_i^\dagger a_j,$$

and

$$\hat{O}_2 = \frac{1}{2} \sum_{ijkl} \langle ij | kl \rangle a_i^\dagger a_j^\dagger a_l a_k,$$

where $\langle i | \hat{h} | j \rangle = \int d\mathbf{x}_1 \chi_i^*(\mathbf{x}_1) \hat{h}(\mathbf{r}_1) \chi_j(\mathbf{x}_2)$ are one-electron integrals while $\langle ij | kl \rangle = \int d\mathbf{x}_1 d\mathbf{x}_2 \chi_i^*(\mathbf{x}_1) \chi_j^*(\mathbf{x}_2) \hat{v}(\mathbf{r}_1, \mathbf{r}_2) \chi_k(\mathbf{x}_1) \chi_l(\mathbf{x}_2)$ are two-electron integrals.

The advantage of utilizing the second quantization formalism is seen when evaluating expectation values of these operators. Given a linear combination of Slater determinants

$$|\psi\rangle = \sum_I C_I |I\rangle,$$

the calculation of the expectation value $\langle \psi | \hat{H} | \psi \rangle = \langle \psi | \hat{O}_1 + \hat{O}_2 | \psi \rangle$ involves evaluating elements such as $\langle I | a_i^\dagger a_j | J \rangle$ and $\langle I | a_i^\dagger a_j^\dagger a_l a_k | J \rangle$. For any number of particles and spin orbitals, these elements can be evaluated using the algebraic properties of the annihilation and creation operators. This results in rules that significantly streamline the process of calculating matrix elements between many-body states. For example, $\langle I | a_i^\dagger a_j | J \rangle$ will be zero for all $|I\rangle$ and $|J\rangle$ differing by more than one spin-orbital. This is due to the orthonormality of the Slater determinants and that the operations $a_i^\dagger a_j$ are only allowing the replacement of one spin-orbital with another.

2.6 Multi-configuration Time-Dependent Hartree-Fock

Our aim of this section is to explain how to approximate the solution for the time-independent Schrödinger equation,

$$\hat{H}(t) |\Psi(t)\rangle = i \frac{\partial}{\partial t} |\Psi(t)\rangle, \quad (7)$$

using the Multi-configuration Time-Dependent Hartree-Fock (MCTDHF) method. In Section 2.4, we showed how to use the full configuration interaction (FCI) method to approximate the eigenstates of a time-independent Hamiltonian. For that purpose, we utilized a time-independent ansatz for the j 'th eigenstate. That is

$$|\Psi^{(j)}\rangle = \sum_I C_I^{(j)} |I\rangle,$$

where the sum over I runs through all the Slater determinants $|I\rangle$ in our basis. For MCTHF however, we will allow both the configuration interaction coefficients, C_I , and the Slater determinants to be time-dependent,

$$|\Psi(t)\rangle = \sum_{1 \leq i_1 < i_2 < \dots < i_n \leq 2M} C_{i_1, i_2, \dots, i_n}(t) |\phi_{i_1}(t) \phi_{i_2}(t) \dots \phi_{i_n}(t)\rangle. \quad (8)$$

We have restricted the number of time-dependent orbitals $\phi_i(t)$ to $2M$ to indicate that we can have a smaller number of time-dependent orbitals than the number of time-independent basis functions K .

To construct the time-dependent orbitals $\phi_i(t)$, we first define an orthonormal basis of spatial orbitals, $\{\psi_1(\mathbf{r}), \psi_2(\mathbf{r}), \dots, \psi_K(\mathbf{r})\}$. A linear combination of these defines M time-dependent spatial orbitals,

$$\phi_i(\mathbf{r}, t) = \sum_{j=1}^K b_{ij}(t) \psi_j(\mathbf{r}), \quad (9)$$

where the orbital matrix has dimensionality $b_{ij} \in \mathbb{C}^{M \times K}$. We will utilize these spatial orbitals to form what is known as a spin-restricted ansatz. Thus we only allow the same set of spatial orbitals $\{\phi_i\}$ and Slater determinants $\{|\phi_{i_1}(t)\phi_{i_2}(t) \cdots \phi_{i_N}(t)\rangle\}$ for the spin α and β subspace. This allows us to write the ansatz in Eq. 8 as a product of Slater determinants for the different spin subspaces, that is

$$|\Psi(t)\rangle = \sum_{I_\alpha, I_\beta} C_{I_\alpha, I_\beta}(t) |I_\alpha(t)\rangle |I_\beta(t)\rangle. \quad (10)$$

Our aim is to approximate the solution for Eq. 7 by varying the parameters C_{I_α, I_β} and b_{ij} . This leads to a set of equations that will be presented in the next sections.

2.6.1 MCTDHF Equations

We start by writing the Hamiltonian of our system in second quantized form, that is,

$$\hat{H}(t) = \sum_{pq, \sigma} h_{pq}(t) a_{p\sigma}^\dagger a_{q\sigma} + \frac{1}{2} \sum_{pqrs, \sigma\tau} g_{pqrs} a_{p\sigma}^\dagger a_{q\tau}^\dagger a_{s\tau} a_{r\sigma}, \quad (11)$$

Where $h_{pq}(t) = \int d\mathbf{r}_1 \phi_p^*(\mathbf{r}_1, t) \hat{h}(t) \phi_q(\mathbf{r}_1, t)$ are the one-body matrix elements and $g_{pqrs} = \int d\mathbf{r}_1 d\mathbf{r}_2 \phi_p^*(\mathbf{r}_1, t) \phi_q^*(\mathbf{r}_2, t) \hat{v}(\mathbf{r}_1, \mathbf{r}_2) \phi_r(\mathbf{r}_1, t) \phi_s(\mathbf{r}_2, t)$ are the two-body matrix elements. The orbitals in Eq. 9 are in general not orthonormal, hence we want to introduce an orthogonality constraint. The problem can then be written in terms of an action functional,

$$S[C_{i_1, \dots, i_N}, \langle \phi_p(t) |] = \int dt \{ \langle \Psi(t) | \hat{H}(t) - i \frac{\partial}{\partial t} | \Psi(t) \rangle - \sum_{kl} \mu_{kl} (\langle \phi_k | \phi_l \rangle - \delta_{kl}) \}. \quad (12)$$

This action functional will need to be minimized in terms of the coefficients, $C_{i_1, \dots, i_N}$, and the spin-orbitals $|\phi_p(t)\rangle$, which will lead to two equations. The minimization will require some knowledge of functional derivatives, a theme that is briefly discussed in appendix D. First we will deal with the equation for the orbital coefficients $b_{ij}(t)$.

2.6.2 Orbital Equation

To find the equation for the orbitals we are interested in finding the orbitals for which

$$\frac{S[C_{i_1, \dots, i_N}, \langle \phi_p(t) |]}{\partial \langle \phi_p(t) |},$$

is stationary, that is, equates to zero. We begin by rewriting the expectation value in the action functional (Eq. 12) by insertion of the second quantized Hamiltonian in Eq. 11,

$$\langle \Psi(t) | \hat{H}(t) - i \frac{\partial}{\partial t} | \Psi(t) \rangle = \sum_{pq} D_{pq} [h_{pq}(t) - (i \frac{\partial}{\partial t})_{pq}] + \frac{1}{2} \sum_{pqrs} g_{pqrs} d_{pqrs},$$

where

$$D_{pq} = D_{pq}^+ + D_{pq}^- = \langle \Psi(t) | a_{p+}^\dagger a_{q+} | \Psi(t) \rangle + \langle \Psi(t) | a_{p-}^\dagger a_{q-} | \Psi(t) \rangle, \quad (13)$$

$$d_{pqrs} = \sum_{\sigma\tau} d_{pqrs}^{\sigma\tau} = \sum_{\sigma\tau} \langle \Psi(t) | a_{p\sigma}^\dagger a_{q\tau}^\dagger a_{s\tau} a_{r\sigma} | \Psi(t) \rangle, \quad (14)$$

$$h_{pq} = \langle \phi_p(\mathbf{r}, t) | \hat{h} | \phi_q(\mathbf{r}, t) \rangle = \int d\mathbf{r}_1 \phi_p^*(\mathbf{r}_1, t) \hat{h}(t) \phi_q(\mathbf{r}_1, t),$$

$$g_{pqrs} = \langle \phi_p(\mathbf{r}, t) \phi_q(\mathbf{r}, t) | \phi_r(\mathbf{r}, t) \phi_s(\mathbf{r}, t) \rangle = \int d\mathbf{r}_1 d\mathbf{r}_2 \phi_p^*(\mathbf{r}_1, t) \phi_q^*(\mathbf{r}_2, t) \hat{v}(\mathbf{r}_1, \mathbf{r}_2) \phi_r(\mathbf{r}_1, t) \phi_s(\mathbf{r}_2, t),$$

$$(i \frac{\partial}{\partial t})_{pq} = i \int d\mathbf{r} \phi_p^*(\mathbf{r}, t) \frac{\partial}{\partial t} \phi_q(\mathbf{r}, t).$$

In order to evaluate the action integral, we will need to apply Eq. 27 together with the product rule of Eq. 28, to the action functional given in Eq. 12. Term by term we have,

$$\frac{\delta(\int dt \sum_{pq} D_{pq} h_{pq})}{\delta \langle \phi_p |} = \sum_q D_{pq} \hat{h} |\phi_q\rangle,$$

where we have utilized the fact that

$$\frac{\delta(\int dt D_{pq})}{\delta \langle \phi_p |} = 0,$$

because of the orthogonality of the Slater determinants. The derivative of the density matrix d_{pqrs} will also be zero for the same reason. We thus have

$$\frac{\delta(\int dt \sum_{pqrs} d_{pqrs} g_{pqrs})}{\delta \langle \phi_p |} = 2 \sum_{qrs} d_{pqrs} \hat{g}_{qs} |\phi_r\rangle,$$

where the factor of 2 comes from the fact that we get a contribution from both particles. We have introduced the mean field operator $\hat{g}_{qs}$ given by

$$g_{qs}(\mathbf{r}) = \int d\mathbf{x} \phi_q^*(\mathbf{x}) v(\mathbf{x}, \mathbf{r}) \phi_s(\mathbf{x}).$$

We also have that

$$\frac{\delta(\int dt \sum_{pq} D_{pq} i(\frac{\partial}{\partial t})_{pq})}{\delta \langle \phi_p |} = \sum_q D_{pq} i \frac{\partial}{\partial t} |\phi_q\rangle.$$

Finally, for the orthogonality constraint we get

$$\frac{\delta(\int dt \sum_{kl} \mu_{kl} (\langle \phi_k | \phi_l \rangle - \delta_{kl}))}{\delta \langle \phi_p |} = \sum_l \mu_{pl} |\phi_l\rangle.$$

Putting this together gives us the functional derivative

$$\begin{aligned} \frac{\delta S[C_I, |\phi_p\rangle]}{\delta |\phi_p\rangle} &= 0 \\ &= \sum_q D_{pq} (\hat{h} - i \frac{\partial}{\partial t}) |\phi_q\rangle + \sum_{qrs} d_{pqrs} \hat{g}_{qs} |\phi_r\rangle - \sum_l \mu_{pl} |\phi_l\rangle. \end{aligned} \tag{15}$$

Projecting onto $\langle \phi_l |$ gives us the solution for the Lagrange multipliers

$$\mu_{pl} = \sum_q D_{pq} \langle \phi_l | (\hat{h} - i \frac{\partial}{\partial t}) |\phi_q\rangle + \sum_{qrs} d_{pqrs} \langle \phi_l | \hat{g}_{qs} |\phi_r\rangle.$$

Inserting this back into Eq. 15 gives

$$0 = \hat{P} (\sum_q D_{pq} (\hat{h} - i \frac{\partial}{\partial t}) |\phi_q\rangle + \sum_{qrs} d_{pqrs} \hat{g}_{qs} |\phi_r\rangle),$$

where

$$\hat{P} = \hat{I} - \sum_l |\phi_l\rangle \langle \phi_l|.$$

Solving for the term with the time-derivative gives

$$i \hat{P} \sum_q D_{pq} \frac{\partial}{\partial t} |\phi_q\rangle = \hat{P} (\sum_q D_{pq} \hat{h} |\phi_q\rangle + \sum_{qrs} d_{pqrs} \hat{g}_{qs} |\phi_r\rangle).$$

We now multiply with $(D^{-1})_{np}$ and sum over p . This gives

$$i \hat{P} \frac{\partial}{\partial t} |\phi_n\rangle = \hat{P} [\hat{h} |\phi_n\rangle + \sum_{pqrs} (D^{-1})_{np} d_{pqrs} \hat{g}_{qs} |\phi_r\rangle]. \tag{16}$$

For any rotation of the MCTDHF orbitals, there exists a corresponding rotation for the coefficients $C_{I\alpha, I\beta}$ such that the wave function becomes identical[4]. This allows us to apply a unitary transformation to the orbitals in order to satisfy

$$\langle \phi_p | i \frac{\partial}{\partial t} | \phi_q \rangle = \langle \phi_p | \hat{Q}(t) | \phi_q \rangle, \quad (17)$$

where $Q(t)$ is a Hermitian operator. The choice of $Q(t)$ is thus arbitrary, but has numerical implications which will not be discussed in this project. Using this in Eq. 16 gives the spin-restricted orbital equation

$$i \frac{\partial}{\partial t} |\phi_n\rangle = \hat{Q} |\phi_n\rangle + \hat{P}[(\hat{h} - \hat{Q}) |\phi_n\rangle + \sum_{pqrs} (D^{-1})_{np} d_{pqrs} \hat{g}_{qs} |\phi_r\rangle].$$

Setting $\hat{Q}(t) = 0$ gives

$$i \frac{\partial}{\partial t} |\phi_n\rangle = \hat{P}[\hat{h} |\phi_n\rangle + \sum_{pqrs} (D^{-1})_{np} d_{pqrs} \hat{g}_{qs} |\phi_r\rangle].$$

We can insert for the time-dependent spatial-orbitals (Eq. 9) and left-project with the spatial orbitals of the original basis, $\langle \psi_i |$. This gives

$$\begin{aligned} i \sum_l \langle \psi_i | \psi_l \rangle \dot{b}_{nl} &= [\langle \psi_i | - \sum_m \sum_{uv} b_{mu} b_{mv}^* \langle \psi_i | \psi_u \rangle \langle \psi_v |] \\ &\cdot [\sum_j b_{nj} \hat{h} |\psi_j\rangle + \sum_{pqrs} (D^{-1})_{np} d_{pqrs} \sum_{jkl} b_{qj}^* b_{sk} b_{rl} \langle \psi_j | \hat{g} | \psi_k \rangle |\psi_l\rangle] \\ &= \sum_j b_{nj} \langle \psi_i | \hat{h} |\psi_j\rangle - \sum_{muvj} b_{mu} b_{mv}^* b_{nj} \langle \psi_i | \psi_u \rangle \langle \psi_v | \hat{h} |\psi_j\rangle \\ &\quad + \sum_{pqrs} (D^{-1})_{np} d_{pqrs} \sum_{jkl} b_{qj}^* b_{sk} b_{rl} \langle \psi_i \psi_j | \hat{g} | \psi_l \psi_k \rangle \\ &\quad - \sum_{pqrs} (D^{-1})_{np} d_{pqrs} \sum_{jklmuv} \langle \psi_i | \psi_u \rangle b_{mu} b_{mv}^* b_{qj}^* b_{sk} b_{rl} \langle \psi_v \psi_j | \hat{g} | \psi_l \psi_k \rangle \\ &= h_{in}^{(1)} - \sum_u S_{iu} h_{un}^{(3)} + \sum_{pqrs} (D^{-1})_{np} d_{pqrs} [g_{iqrs}^{(3)} - \sum_u S_{iu} g_{uqrs}^{(5)}], \end{aligned}$$

that is

$$i \sum_l S_{il} \dot{b}_{nl} = h_{in}^{(1)} - \sum_u S_{iu} h_{un}^{(3)} + \sum_{pqrs} (D^{-1})_{np} d_{pqrs} [g_{iqrs}^{(3)} - \sum_u S_{iu} g_{uqrs}^{(5)}],$$

where S is the overlap matrix for the time-independent spatial orbitals. We have used the following definitions for the transformed one-body and two-body operators,

$$h_{jk}^{(0)} = \langle \psi_j | \hat{h} | \psi_k \rangle.$$

$$h_{jn}^{(1)} = \langle \psi_j | \hat{h} | \phi_n \rangle = \sum_{k=1}^{2K} b_{nk} h_{jk}^{(0)}.$$

$$h_{mn} = \langle \phi_m | \hat{h} | \phi_n \rangle = \sum_{j=1}^{2K} b_{mj}^* h_{jn}^{(1)}.$$

$$h_{jn}^{(3)} = \sum_{m=1}^M \langle \psi_j | \phi_m \rangle \langle \phi_m | \hat{h} | \phi_n \rangle = \sum_{m=1}^M b_{mj} h_{mn}$$

$$g_{ijkl}^{(0)} = \langle \psi_i \psi_j | \psi_k \psi_l \rangle.$$

$$g_{iqks}^{(2)} = \sum_{j=1}^{2K} b_{qj}^* \sum_{l=1}^{2K} b_{sl} g_{ijkl}^{(0)}.$$

$$g_{i q r s}^{(3)} = \sum_{k=1}^{2K} b_{rk} g_{i q k s}^{(2)}.$$

$$g_{p q r s} = \sum_{i=1}^{2K} b_{pi}^* g_{i q r s}^{(3)}.$$

$$g_{i q r s}^{(5)} = \sum_{p=1}^M b_{pi} g_{p q r s}.$$

As we will be working with an orthonormal time-independent basis, S is diagonal, and the equation reduces to

$$i\dot{b}_{ni} = h_{in}^{(1)} - h_{in}^{(3)} + \sum_{p q r s} (D^{-1})_{np} d_{p q r s} [g_{i q r s}^{(3)} - g_{i q r s}^{(5)}],$$

which is the orbital equation that will be utilized in this project

2.6.3 Coefficient equation

To find the equation for the coefficients, we take the derivative of the time-dependent Schrödinger equation (Eq. 7) wrt. the coefficients. The equation can be written out as follows

$$\langle \Psi(t) | \hat{H}(t) - i \frac{d}{dt} | \Psi(t) \rangle = \sum_{I,J} C_I^* \langle I | (\hat{H}(t) - i \frac{\partial}{\partial t}) C_J | J \rangle = 0$$

The derivative of this equation wrt. C_I^* gives

$$\begin{aligned} \sum_J \langle I | (\hat{H}(t) - i \frac{\partial}{\partial t}) C_J | J \rangle &= \sum_J C_J \langle I | (\hat{H}(t) - i \frac{\partial}{\partial t}) | J \rangle - i \sum_J \frac{\partial C_J}{\partial t} \langle I | J \rangle = 0 \\ \Rightarrow i \frac{\partial C_I}{\partial t} &= \sum_J C_J \langle I | (\hat{H}(t) - i \frac{\partial}{\partial t}) | J \rangle \end{aligned}$$

By using Eq. 17 we get

$$\begin{aligned} i \frac{\partial C_I}{\partial t} &= \sum_J C_J \langle I | (\hat{H}(t) - Q(t)) | J \rangle \\ i \frac{\partial C_I}{\partial t} &= \sum_J C_J \langle I | \hat{H}(t) | J \rangle, \end{aligned} \tag{18}$$

where we set $Q(t) = 0$ in the last step.

2.6.4 Reducing the computational complexity

The spin-restricted MCTDHF ansatz (Eq. 10) consists of products of Slater determinants for the two spin sub-spaces. We can exploit this to reduce the complexity when calculating the one- and two-body density matrices in Eqs. 13 and 14. For the one-body density we have

$$D_{pq}^\sigma = \sum_{I_\alpha I_\beta J_\alpha J_\beta} C_{I_\alpha I_\beta}^* C_{J_\alpha J_\beta} \langle I_\alpha | \langle I_\beta | a_{p\sigma}^\dagger a_{q\sigma} | J_\alpha \rangle | J_\beta \rangle.$$

For the α and β subspace we have two different equations,

$$D_{pq}^\alpha = \sum_{I_\alpha I_\beta J_\alpha J_\beta} \delta_{I_\beta, J_\beta} C_{I_\alpha I_\beta}^* C_{J_\alpha J_\beta} \langle I_\alpha | a_{p\alpha}^\dagger a_{q\alpha} | J_\alpha \rangle,$$

and

$$D_{pq}^\beta = \sum_{I_\alpha I_\beta J_\alpha J_\beta} \delta_{I_\alpha, J_\alpha} C_{I_\alpha I_\beta}^* C_{J_\alpha J_\beta} \langle I_\beta | a_{p\beta}^\dagger a_{q\beta} | J_\beta \rangle,$$

where the $\delta_{I,J}$ comes from the orthonormality of Slater determinants. For the two-body density we get

$$d_{pqrs}^{\sigma\tau} = \sum_{I_\alpha I_\beta J_\alpha J_\beta} C_{I_\alpha I_\beta}^* C_{J_\alpha J_\beta} \langle I_\alpha | \langle I_\beta | a_{p\sigma}^\dagger a_{q\tau}^\dagger a_{s\tau} a_{r\sigma} | J_\alpha \rangle | J_\beta \rangle.$$

For the different spin-indices this becomes

$$d_{pqrs}^{\alpha\alpha} = \sum_{I_\alpha I_\beta J_\alpha J_\beta} \delta_{I_\beta J_\beta} C_{I_\alpha I_\beta}^* C_{J_\alpha J_\beta} \langle I_\alpha | a_{p\alpha}^\dagger a_{q\alpha}^\dagger a_{s\alpha} a_{r\alpha} | J_\alpha \rangle,$$

$$d_{pqrs}^{\beta\beta} = \sum_{I_\alpha I_\beta J_\alpha J_\beta} \delta_{I_\alpha J_\alpha} C_{I_\alpha I_\beta}^* C_{J_\alpha J_\beta} \langle I_\beta | a_{p\beta}^\dagger a_{q\beta}^\dagger a_{s\beta} a_{r\beta} | J_\beta \rangle,$$

$$d_{pqrs}^{\alpha\beta} = \sum_{I_\alpha I_\beta J_\alpha J_\beta} C_{I_\alpha I_\beta}^* C_{J_\alpha J_\beta} \langle I_\alpha | a_{p\alpha}^\dagger a_{r\alpha} | J_\alpha \rangle \langle I_\beta | a_{q\beta}^\dagger a_{s\beta} | J_\beta \rangle,$$

and

$$d_{pqrs}^{\beta\alpha} = \sum_{I_\alpha I_\beta J_\alpha J_\beta} C_{I_\alpha I_\beta}^* C_{J_\alpha J_\beta} \langle I_\alpha | a_{q\alpha}^\dagger a_{s\alpha} | J_\alpha \rangle \langle I_\beta | a_{p\beta}^\dagger a_{r\beta} | J_\beta \rangle.$$

Thus the calculation of the one body densities reduces to calculating the one and two body overlaps, $\langle I_\sigma | a_{p\sigma}^\dagger a_{q\sigma} | J_\sigma \rangle$ and $\langle I_\sigma | a_{p\sigma}^\dagger a_{q\sigma}^\dagger a_{s\sigma} a_{r\sigma} | J_\sigma \rangle$, separately for each spin-subspace. The same procedure can be used for the coefficient equation (Eq. 20). That is

$$\begin{aligned} i \frac{\partial C_{I_\alpha I_\beta}}{\partial t} &= \sum_{J_\alpha} C_{J_\alpha I_\beta} \sum_{pq} h_{pq} \langle I_\alpha | a_{p\alpha}^\dagger a_{q\alpha} | J_\alpha \rangle \\ &+ \sum_{J_\beta} C_{I_\alpha J_\beta} \sum_{pq} h_{pq} \langle I_\beta | a_{p\beta}^\dagger a_{q\beta} | J_\beta \rangle \\ &+ \frac{1}{2} \sum_{J_\alpha} C_{J_\alpha I_\beta} \sum_{pqrs} g_{pqrs} \langle I_\alpha | a_{p\alpha}^\dagger a_{q\alpha}^\dagger a_{s\alpha} a_{r\alpha} | J_\alpha \rangle \\ &+ \frac{1}{2} \sum_{J_\beta} C_{I_\alpha J_\beta} \sum_{pqrs} g_{pqrs} \langle I_\beta | a_{p\beta}^\dagger a_{q\beta}^\dagger a_{s\beta} a_{r\beta} | J_\beta \rangle \\ &+ \frac{1}{2} \sum_{J_\alpha J_\beta} C_{J_\alpha J_\beta} \sum_{pqrs} \langle I_\alpha | a_{p\alpha}^\dagger a_{r\alpha} | J_\alpha \rangle \langle I_\beta | a_{q\beta}^\dagger a_{s\beta} | J_\beta \rangle g_{pqrs} \\ &+ \frac{1}{2} \sum_{J_\alpha J_\beta} C_{J_\alpha J_\beta} \sum_{pqrs} \langle I_\alpha | a_{q\alpha}^\dagger a_{s\alpha} | J_\alpha \rangle \langle I_\beta | a_{p\beta}^\dagger a_{r\beta} | J_\beta \rangle g_{pqrs}. \end{aligned}$$

Again, we are only reliant on calculating the overlaps $\langle I_\sigma | a_{p\sigma}^\dagger a_{q\sigma} | J_\sigma \rangle$ and $\langle I_\sigma | a_{p\sigma}^\dagger a_{q\sigma}^\dagger a_{s\sigma} a_{r\sigma} | J_\sigma \rangle$. These will be identical for each spin-subspace if they contain an equal number of particles.

3 Method

3.1 One- and Two-body Potential and External Field

For all our simulations, the interaction we will consider is a one dimensional Coloumb-interaction given by

$$u(x_i, x_j) = \frac{\kappa}{|x_i - x_j| + \alpha},$$

where x_i and x_j is the position of particle i and j , respectively. The parameter κ decides the strength of the interaction, while α is a screening parameter introduced to avoid division by zero when integration is performed. The one-body potentials we will consider in this project consists of a simple Harmonic oscillator potential,

$$V(x) = \frac{1}{2} \omega^2 x^2,$$

where ω is the angular frequency and x is the position. The time-dependent part of the Hamiltonian is given by an external laser field, which under the dipole approximation can be represented as,

$$v(x, t) = \epsilon_0 x \sin(\omega_L t). \quad (19)$$

Here ϵ_0 is the strength of the field, while ω_L is the angular frequency of the laser.

For all simulations we will utilize the same parameters as in Zanghellini *et al.* [1], that is, $\kappa = 1$, $a = \omega = 0.25$, $\epsilon_0 = 1$ and $\omega_L = 8\omega$.

3.2 Finite Difference Single Particle Basis

The spatial basis states we will utilize in this project is a finite difference basis. We first define our basis states as the vectors,

$$\vec{\psi}_i = \begin{bmatrix} \psi_i(x_0) \\ \psi_i(x_1) \\ \vdots \\ \psi_i(x_n) \end{bmatrix},$$

where $x_{i+1} - x_i = \Delta x$ is the spacing between the points on a uniform and discrete grid. Similarly to Zanghellini *et al.*, we will always utilize a grid from -10 to 10 with a spacing of $\Delta x = 0.1$. We assume the boundary conditions $\psi_i(x_0) = \psi_i(x_n) = 0$. If we write our one-body Hamiltonian as

$$h = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \dots \\ \frac{1}{2(\Delta x)^2} & -\frac{1}{(\Delta x)^2} & \frac{1}{2(\Delta x)^2} & 0 & 0 & 0 & \dots \\ 0 & \frac{1}{2(\Delta x)^2} & -\frac{1}{(\Delta x)^2} & \frac{1}{2(\Delta x)^2} & 0 & 0 & \dots \\ 0 & 0 & \frac{1}{2(\Delta x)^2} & -\frac{1}{(\Delta x)^2} & \frac{1}{2(\Delta x)^2} & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & \dots & 0 & 0 \\ 0 & V(x_1) & 0 & \dots & 0 & 0 \\ 0 & 0 & V(x_2) & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & V(x_{n-1}) & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 \end{bmatrix},$$

we see that the k 'th element $(h\vec{\psi}_i)_k = -\frac{1}{2} \frac{\psi_i(x_{k+1}) - 2\psi_i(x_k) + \psi_i(x_{k-1}))}{(\Delta x)^2} + V(x_k)\psi_i(x_k)$. The factor $\frac{\psi_i(x_{k+1}) - 2\psi_i(x_k) + \psi_i(x_{k-1}))}{(\Delta x)^2}$ corresponds to the second symmetric derivative when $\Delta x \rightarrow 0$, thus we have a discrete representation of the one body Hamiltonian $h(x) = -\frac{1}{2} \frac{\partial^2}{\partial x^2} + V(x)$. Our single particle basis will be the K lowest eigenvalue eigenvectors of this one body Hamiltonian.

3.3 Time Evolution and numerical integration

One is often interested in finding the ground state of a Hamiltonian at an initial time $H(t_0)$ and then evolve of the ground state under the influence of said Hamiltonian. One example is the usage of quantum computers, where the qubits are initialized in their ground state, then external time-dependent fields are applied to rotate and entangle the qubits. The initial state can be found by diagonalization of $H(t_0)$, but this often involves construction of the whole Hamiltonian matrix and the calculation of the complete spectrum. In this project, we will apply the imaginary time evolution to find the ground state. This is done by the substitution of $t \rightarrow -i\tau$ in the coefficient and orbital equation. This leads to the following equations,

$$\dot{b}_{ni} = -(h_{in}^{(1)} - h_{in}^{(3)} + \sum_{pqrs} (D^{-1})_{np} d_{pqrs} [g_{iqrs}^{(3)} - g_{iqrs}^{(5)}]),$$

and

$$\frac{\partial C_I}{\partial t} = - \sum_J C_J \langle I | \hat{H}(t) | J \rangle. \quad (20)$$

The numerical integration for both the real and imaginary time evolution will be performed by the *ode* class from the Scipy [5] sub-package *integrate*. The integration method is called *zvode*, which allows for a dynamic step size in the integration. We will in all simulations use a maximum step size of $1e-2$ and $1e-3$ for imaginary and real time evolution, respectively. For the imaginary time evolution, the CI coefficients, C , will be normalized at each integration step. The orbital coefficients, b , will be orthogonalized with the Gram-Schmidt procedure at each integration step.

4 Results and Discussion

Our aim is to compare our simulations with the results in section 3 of the article by Zanghellini *et al.* [1]. Since they utilize a spin-unrestricted ansatz their state-space is larger than for our spin-restricted ansatz, and direct comparisons of the achieved ground state energies are difficult. We are able to compare the ground state energies for when we utilize one MCTDHF orbital for each spin-subspace, the Hartree-Fock scenario. As we increase the number of orbitals, we compare our results with the (spin-restricted) CI ground state energy, as this is the lowest achievable energy within the subspace spanned by our single particle basis. For all our simulations, we utilize 10 single particle basis functions. The results for imaginary time evolution can be seen in figure 1.

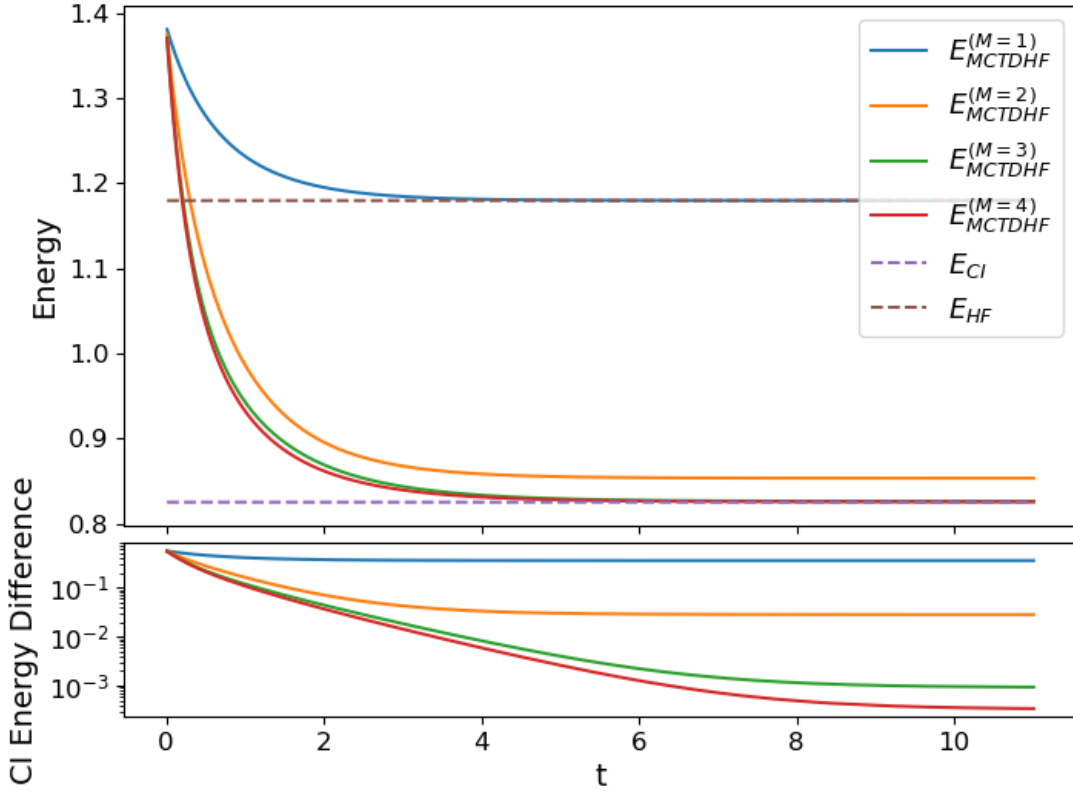


Figure 1: Comparison of the MCTDHF energy with M spatial orbitals within each spin-subspace. The top plot shows the energy against time for imaginary time evolution for different M . We also plot the CI energy, E_{CI} and the results from Zanghellini *et al.* for $M = 1$, denoted E_{HF} . The bottom plot shows the absolute value of the difference between the CI energy and the MCTDHF energies for different M .

We see that for $M = 1$, the MCTDHF energy converges towards the corresponding results from Zanghellini *et al.*, denoted E_{HF} . From the bottom part of the figure, we see that the MCTDHF results converge towards the CI energy as we increase the number of MCTDHF orbitals.

Next we compare the particle density of our ground states against the particle densities achieved by Zanghellini *et al.*. In their work, η corresponds to the number of CI coefficients in their ansatz. The results can be seen in figures 2 and 3. We see that there is a good correspondence between our result and the result of Zanghellini *et al.* for the Hartree-Fock scenario ($M = 1$ and $\eta = 1$). Our results for $M = 3$ and $M = 4$ are also visually identical to the $\eta = \infty$ (exact results) density in the Zanghellini *et al.* article. However, we see that we are not able to achieve their result for $\eta = 6$. We suspect that this might be due to our ansatz being spin-restricted, thus the number of configurations is given by $\eta = \binom{M}{1}\binom{M}{1} = M^2$. We are hence not able to generate an ansatz with $\eta = 6$ configurations.

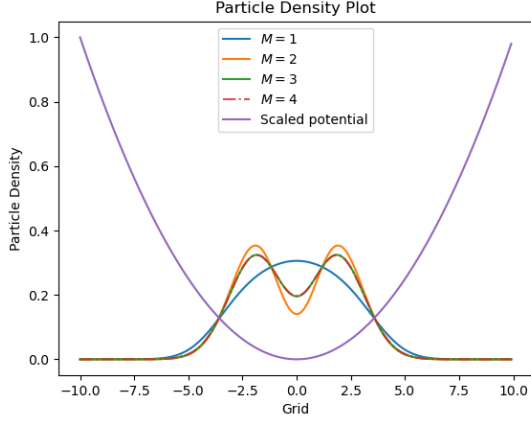


Figure 2: Particle density against the grid (x) for different number of MCTDHF orbitals M . Scaled potential shows the Harmonic oscillator potential, but divided by the maximum value over the grid.

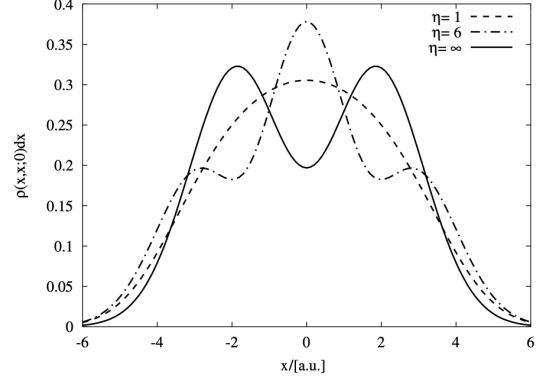


Figure 3: Results of Zanghellini *et al.* for the harmonic quantum dot. Figure and description taken from their article[1]: Electron density $\rho(x, x; 0)$ for the ground-state wavefunction and an increasing number of configurations η . ($\eta = \infty$ refers to the numerically exact solution.) The result for $\eta = 15$ coincides with the exact result within the resolution of the plot.

Finally, we will compare the real time evolution of our ground states subject to the monochromatic laser field given by eq. 19. We again compare our simulations with the ones from Zanghellini *et al.*. This entails performing real time evolution and calculating the overlap between the ground state at time 0 to the state at a later time t , $|\langle \Psi(0) | \Psi(t) \rangle|^2$. The results can be seen in figures 4 and 5. We see that there is a good correspondence between the results for $M = 1$ and $\eta = 1$, which is to be expected. For $M = 3$ and $M = 4$, we see that our results are visually identical to the exact solution ($\eta = \infty$) from Zanghellini *et al.*.

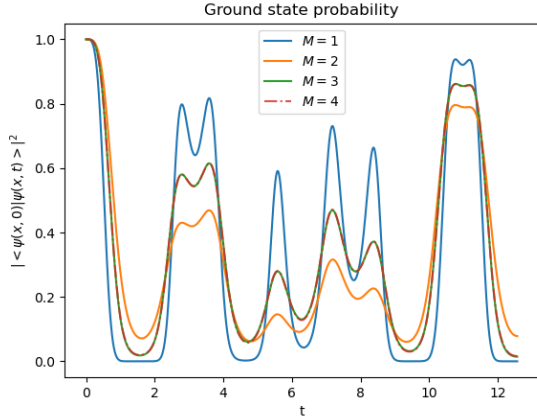


Figure 4: Probability of being in the ground state $|\langle \Psi_M(x_1, x_2; 0) | \Psi_M(x_1, x_2; t) \rangle|^2$ for different number of MCTDHF orbitals M . The results for $M = 3$ and $M = 4$ are visually identical.

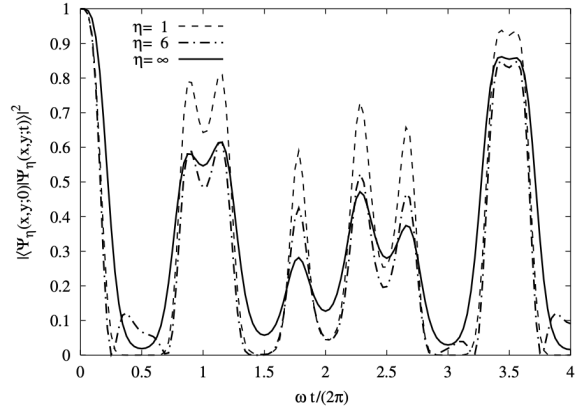


Figure 5: Results of Zanghellini *et al.* for the harmonic quantum dot. Figure and description taken from their article[1]: Probability of being in the ground state $|\langle \Psi_\eta(x, y; 0) | \Psi_\eta(x, y; t) \rangle|^2$ for an increasing number of configurations η ($\eta = \infty$ refers to the numerically exact solution) in the presence of an electric field of strength $\epsilon_0 = 1$ and frequency $\omega_L = 8\omega$. The result for $\eta = 15$ coincides with the exact result within the resolution of the plot.

5 Conclusion

In this project, we have implemented the spin-restricted Multi-Configurational Time-Dependent Hartree-Fock method and investigated the time evolution of two electrons in a quantum harmonic dot potential. The results were compared with the work by Zanghellini *et al.* [1]. Although they utilized a spin-unrestricted variant of the algorithm, we found a good correspondence when only utilizing one MCTDHF orbital for each spin-subspace and when our system got close to the optimal ground state energy (calculated from the spin-restricted configuration interaction method). In these scenarios, both the particle densities and the time evolution of our states visually coincided with the results of Zanghellini *et al.*. However, we found that an exact replication of their results was difficult due to the different way of handling spin in our work.

5.1 Further Work

When continuing the investigation of the MCTDHF method, we would like to further optimize the algorithm. This entails implementing direct time dependent configuration interaction (see section 3.4.2 in [4]), also known as the minimal operation-count method. Utilizing Python packages for high performance computing such as *jax* [6] is also planned in order to achieve speedups. Further, we would also like to investigate the utilization of basis sets other than the finite difference basis. An example is one of the discrete variable representation (DVR) basis sets (see appendix B.1 in [4]). Finally we would like to allow for two- and three-dimensional systems and also the spin-unrestricted MCTDHF ansatz. The latter would be the ideal choice for spin-dependent systems.

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A Orthonormality of Slater determinants

In this section, we show that the N -particle Slater determinants consisting of orthonormal basis functions $\{\chi_i\}$ are orthonormal. To do this, we consider an alternative way of writing a Slater determinant, that is

$$|\chi_i \chi_j \cdots \chi_k\rangle = \frac{1}{\sqrt{N!}} \sum_{n=1}^{N!} (-1)^{p_n} \hat{P}_n \{\chi_i(\mathbf{x}_1) \chi_j(\mathbf{x}_2) \cdots \chi_k(\mathbf{x}_N)\}, \quad (21)$$

where $\hat{P}_n$ generates the n 'th permutation of the electrons and p_n is the number of simple interchanges that is required to perform the permutation. For example, one possible permutation is $\hat{P}_q \{\chi_i(\mathbf{x}_1) \chi_j(\mathbf{x}_2) \cdots \chi_k(\mathbf{x}_N)\} =$

$\chi_i(\mathbf{x}_2)\chi_j(\mathbf{x}_1)\cdots\chi_k(\mathbf{x}_N)$, which requires $p_q = 1$ simple interchange. Given two Slater determinants $|K\rangle = |\chi_i\chi_j\cdots\chi_k\rangle$ and $|J\rangle = |\chi_{i'}\chi_{j'}\cdots\chi_{k'}\rangle$, for which we assume the columns have been ordered for optimal matching, their inner product becomes

$$\frac{1}{N!} \sum_{n=1}^{N!} \sum_{m=1}^{N!} (-1)^{p_n} (-1)^{p_m} \int d\mathbf{x}_1 d\mathbf{x}_2 \cdots d\mathbf{x}_N \hat{P}_n \{\chi_i^*(\mathbf{x}_1)\chi_j^*(\mathbf{x}_2)\cdots\chi_k^*(\mathbf{x}_N)\} \hat{P}_m \{\chi_{i'}(\mathbf{x}_1)\chi_{j'}(\mathbf{x}_2)\cdots\chi_{k'}(\mathbf{x}_N)\}.$$

If the above term equates to a non-zero value, the primed and unprimed orbitals must be identical. If they were not, the integration of $\int d\mathbf{x}_l \chi_q^*(\mathbf{x}_l)\chi_{q'}(\mathbf{x}_l)$ for some $q \neq q'$ will equate to zero because of the orthogonality of the spin orbitals. Hence $\langle K|J\rangle = 0$ for $|K\rangle \neq |J\rangle$. For $|K\rangle = |J\rangle$, we get contributions only when the permutation of the primed and unprimed orbitals are the same, that is when $n = m$. When the permutations are not the same, we will again end up with a factor of $\int d\mathbf{x}_l \chi_q^*(\mathbf{x}_l)\chi_{q'}(\mathbf{x}_l)$ for some $q \neq q'$, which will equate to zero. Since $(-1)^{2p_n} = 1$, we then have

$$\frac{1}{N!} \sum_{n=1}^{N!} \int d\mathbf{x}_1 d\mathbf{x}_2 \cdots d\mathbf{x}_N \hat{P}_n \{\chi_i^*(\mathbf{x}_1)\chi_j^*(\mathbf{x}_2)\cdots\chi_k^*(\mathbf{x}_N)\} \hat{P}_n \{\chi_i(\mathbf{x}_1)\chi_j(\mathbf{x}_2)\cdots\chi_k(\mathbf{x}_N)\} = \frac{1}{N!} \sum_{n=1}^{N!} = 1.$$

Correspondingly, we have the final result

$$\langle K|J\rangle = \delta_{KJ}.$$

B Matrix Elements

With our Hamiltonian written in terms of a sum of one-electron operators

$$\hat{O}_1 = \sum_{i=1}^N \hat{h}(\mathbf{r}_i), \quad (22)$$

and a sum of two-electron operators

$$\hat{O}_2 = \sum_{i=1}^N \sum_{j>i}^N \hat{v}(\mathbf{r}_i, \mathbf{r}_j), \quad (23)$$

we are interested in evaluating the matrix elements $\langle K|\hat{O}|J\rangle$ for $|K\rangle = |\cdots ij\cdots\rangle$ and $|J\rangle = |\cdots kl\cdots\rangle$. We will consider three different scenarios for the states $|K\rangle$ and $|J\rangle$. For the first scenario, $|K\rangle = |J\rangle$. We thus evaluate $\langle K|\hat{O}|K\rangle$ for

$$|K\rangle = |\cdots ij\cdots\rangle.$$

In the second scenario the Slater determinants are differing by one spin orbital, that is

$$|K\rangle = |\cdots ij\cdots\rangle \quad \text{and} \quad |J\rangle = |\cdots ik\cdots\rangle.$$

The final scenario concerns when the Slater determinants are differing by two spin orbitals, that is

$$|K\rangle = |\cdots ij\cdots\rangle \quad \text{and} \quad |J\rangle = |\cdots kl\cdots\rangle.$$

Fortunately, we will see that the matrix elements become zero for all other scenarios.

We will start by dealing with one of the terms in the one-electron operator in Eq. 22. By writing the Slater determinants on the form in Eq. 21, we have for the first scenario

$$\begin{aligned} \langle K|\hat{h}(\mathbf{r}_1)|K\rangle &= \frac{1}{N!} \sum_{n,m=1}^{N!} (-1)^{p_n} (-1)^{p_m} \\ &\cdot \int d\mathbf{x}_1 d\mathbf{x}_2 \cdots d\mathbf{x}_N \hat{P}_n \{\chi_i^*(\mathbf{x}_1)\chi_j^*(\mathbf{x}_2)\cdots\chi_k^*(\mathbf{x}_N)\} \hat{h}(\mathbf{r}_1) \hat{P}_m \{\chi_i(\mathbf{x}_1)\chi_j(\mathbf{x}_2)\cdots\chi_k(\mathbf{x}_N)\}. \end{aligned}$$

First off, we can see that the permutations will have to be equal, $m = n$, for a term to give a non-zero value. This is due to the orthogonality of the spin orbitals. We thus have

$$\begin{aligned} \langle K|\hat{h}(\mathbf{r}_1)|K\rangle &= \frac{1}{N!} \sum_n^{N!} \\ &\cdot \int d\mathbf{x}_1 d\mathbf{x}_2 \cdots d\mathbf{x}_N \hat{P}_n \{\chi_i^*(\mathbf{x}_1)\chi_j^*(\mathbf{x}_2)\cdots\chi_k^*(\mathbf{x}_N)\} \hat{h}(\mathbf{r}_1) \hat{P}_n \{\chi_i(\mathbf{x}_1)\chi_j(\mathbf{x}_2)\cdots\chi_k(\mathbf{x}_N)\}. \end{aligned}$$

Secondly, we can see that the integrals involving other integration variables than $\mathbf{x}_1$ will evaluate to one, since they involve the inner product of two identical spin orbitals. The integration variable $\mathbf{x}_1$ will be involved in each spin orbital $(N - 1)!$ times, since this is the number of ways to arrange the other electrons in the remaining $N - 1$ spin orbitals. We thus have

$$\begin{aligned} \langle K | \hat{h}(\mathbf{r}_1) | K \rangle &= \frac{1}{N} \sum_m^N \\ &\cdot \int d\mathbf{x}_1 \chi_m^*(\mathbf{x}_1) \hat{h}(\mathbf{r}_1) \chi_m(\mathbf{x}_1) = \frac{1}{N} \sum_m^N \langle m | \hat{h}(\mathbf{r}_1) | m \rangle. \end{aligned}$$

This result shows that the expectation value is indifferent as to which electron the operator is acting upon. This is due to electrons being indistinguishable particles, and will thus be applicable to all of our following derivations. Hence, for the full one-body operator in Eq. 22, we have

$$\langle K | \hat{O}_1 | K \rangle = \sum_m^N \langle m | \hat{h} | m \rangle.$$

For the second scenario, we will evaluate $\langle ij \dots | \hat{h}(\mathbf{r}_1) | qj \dots \rangle$. That is

$$\begin{aligned} \langle K | \hat{h}(\mathbf{r}_1) | J \rangle &= \frac{1}{N!} \sum_{n,m=1}^{N!} (-1)^{p_n} (-1)^{p_m} \\ &\cdot \int d\mathbf{x}_1 d\mathbf{x}_2 \dots d\mathbf{x}_N \hat{P}_n \{ \chi_i^*(\mathbf{x}_1) \chi_j^*(\mathbf{x}_2) \dots \chi_k^*(\mathbf{x}_N) \} \hat{h}(\mathbf{r}_1) \hat{P}_m \{ \chi_q(\mathbf{x}_1) \chi_j(\mathbf{x}_2) \dots \chi_k(\mathbf{x}_N) \}. \end{aligned}$$

First of all, since spin orbital q and i is orthogonal to every other spin orbital, they will have to be occupied with electron 1 to "match" $\hat{h}(\mathbf{r}_1)$ and give a non-zero term. Following the arguments for the previous scenario, we only get non-zero when the n 'th and m 'th permutation is the same for the remaining $N - 1$ electrons. The other $N - 1$ electrons have $(N - 1)!$ different ways to be arranged. We thus have

$$\langle K | \hat{h}(\mathbf{r}_1) | J \rangle = \frac{1}{N} \cdot \int d\mathbf{x}_1 \chi_i^*(\mathbf{x}_1) \hat{h}(\mathbf{r}_1) \chi_q(\mathbf{x}_1) = \frac{1}{N} \langle i | \hat{h}(\mathbf{r}_1) | q \rangle.$$

Summing all the terms in the one-electron operator gives

$$\langle ij \dots | \hat{O}_1 | iq \dots \rangle = \langle i | \hat{h} | q \rangle.$$

For the final scenario we have

$$\langle ij \dots | \hat{O}_1 | qr \dots \rangle = 0.$$

This can be seen from

$$\begin{aligned} \langle K | \hat{h}(\mathbf{r}_1) | J \rangle &= \frac{1}{N!} \sum_{n,m=1}^{N!} (-1)^{p_n} (-1)^{p_m} \\ &\cdot \int d\mathbf{x}_1 d\mathbf{x}_2 \dots d\mathbf{x}_N \hat{P}_n \{ \chi_i^*(\mathbf{x}_1) \chi_j^*(\mathbf{x}_2) \dots \chi_k^*(\mathbf{x}_N) \} \hat{h}(\mathbf{r}_1) \hat{P}_m \{ \chi_q(\mathbf{x}_1) \chi_r(\mathbf{x}_2) \dots \chi_k(\mathbf{x}_N) \}, \end{aligned}$$

since there are no permutation that avoids an integral between orthogonal spin orbitals. For all other scenarios, we will get a zero for the same reason.

For the two-electron operator, we will again start with one of the terms in Eq. 23 for the first scenario. That is

$$\begin{aligned} \langle K | \hat{v}(\mathbf{r}_1, \mathbf{r}_2) | J \rangle &= \frac{1}{N!} \sum_{n,m=1}^{N!} (-1)^{p_n} (-1)^{p_m} \\ &\cdot \int d\mathbf{x}_1 d\mathbf{x}_2 \dots d\mathbf{x}_N \hat{P}_n \{ \chi_i^*(\mathbf{x}_1) \chi_j^*(\mathbf{x}_2) \dots \chi_k^*(\mathbf{x}_N) \} \hat{v}(\mathbf{r}_1, \mathbf{r}_2) \hat{P}_m \{ \chi_i(\mathbf{x}_1) \chi_j(\mathbf{x}_2) \dots \chi_k(\mathbf{x}_N) \}. \end{aligned}$$

Firstly, note that electron 3, 4, $\dots$, N must occupy the same spin orbitals in both permutations for a term to give a non-zero value due to the orthonormality of the spin orbitals. The remaining two electrons must thus occupy the

remaining two spin orbitals in both permutations. If the left permutation is $\chi_i(\mathbf{r}_1)\chi_j(\mathbf{r}_2)\cdots$, we have the possibility of $\chi_i(\mathbf{r}_1)\chi_j(\mathbf{r}_2)\cdots$ and $\chi_i(\mathbf{r}_2)\chi_j(\mathbf{r}_1)\cdots$ in the right permutation. In either case, there is a possibility of attaining a non-zero term. We can represent the interchange of these two electrons with the operator $\hat{P}_{12}$. That is,

$$\hat{P}_{12}\{\chi_i(\mathbf{r}_1)\chi_j(\mathbf{r}_2)\cdots\} = -\chi_i(\mathbf{r}_2)\chi_j(\mathbf{r}_1)\cdots.$$

This allows us to rewrite our matrix element as

$$\begin{aligned} \langle K|\hat{v}(\mathbf{r}_1, \mathbf{r}_2)|J\rangle &= \frac{1}{N!} \sum_n^{N!} \\ &\cdot \int d\mathbf{x}_1 d\mathbf{x}_2 \cdots d\mathbf{x}_N \hat{P}_n \{\chi_i^*(\mathbf{x}_1)\chi_j^*(\mathbf{x}_2)\cdots\chi_k^*(\mathbf{x}_N)\} \hat{v}(\mathbf{r}_1, \mathbf{r}_2) (1 - \hat{P}_{12}) \hat{P}_n \{\chi_i(\mathbf{x}_1)\chi_j(\mathbf{x}_2)\cdots\chi_k(\mathbf{x}_N)\}. \end{aligned}$$

The integrals involving electrons 3, 4, $\dots$, N will evaluate to one due to the orthonormality of the spin orbitals. The number of ways to arrange these $N - 2$ electrons hence leads to a factor of $(N - 2)!$. Since electron 1 and 2 can occupy any two spin orbitals, we get

$$\begin{aligned} \langle K|\hat{v}(\mathbf{r}_1, \mathbf{r}_2)|J\rangle &= \frac{1}{N(N - 1)} \sum_m^N \sum_{n \neq m}^N \\ &\cdot \int d\mathbf{x}_1 d\mathbf{x}_2 \chi_n^*(\mathbf{x}_1)\chi_m^*(\mathbf{x}_2) \hat{v}(\mathbf{r}_1, \mathbf{r}_2) (1 - \hat{P}_{12}) \chi_n(\mathbf{x}_1)\chi_m(\mathbf{x}_2) \\ &= \frac{1}{N(N - 1)} \sum_m^N \sum_{n \neq m}^N (\langle nm|\hat{v}(\mathbf{r}_1, \mathbf{r}_2)|nm\rangle - \langle nm|\hat{v}(\mathbf{r}_1, \mathbf{r}_2)|mn\rangle). \end{aligned}$$

Since the term when $n = m$ vanishes we can remove the restriction on the sum. The above integral is not dependent on the electron labels, so we can evaluate the matrix element of the full two-electron operator in Eq. 23 by introducing the appropriate factor. Since there are $N(N - 1)/2$ terms in the operator, we get

$$\langle K|\hat{O}_2|K\rangle = \frac{1}{2} \sum_n^N \sum_m^N (\langle nm|\hat{v}|nm\rangle - \langle nm|\hat{v}|mn\rangle).$$

For the second scenario, we want to evaluate

$$\begin{aligned} \langle ij \cdots |\hat{v}(\mathbf{r}_1, \mathbf{r}_2)|kj \cdots\rangle &= \frac{1}{N!} \sum_{n,m=1}^{N!} (-1)^{p_n} (-1)^{p_m} \\ &\cdot \int d\mathbf{x}_1 d\mathbf{x}_2 \cdots d\mathbf{x}_N \hat{P}_n \{\chi_i^*(\mathbf{x}_1)\chi_j^*(\mathbf{x}_2)\cdots\} \hat{v}(\mathbf{r}_1, \mathbf{r}_2) \hat{P}_m \{\chi_k(\mathbf{x}_1)\chi_j(\mathbf{x}_2)\cdots\}. \end{aligned}$$

Following the same arguments as for the previous case, we can rewrite this as

$$\begin{aligned} \langle K|\hat{v}(\mathbf{r}_1, \mathbf{r}_2)|J\rangle &= \frac{1}{N!} \sum_n^{N!} \\ &\cdot \int d\mathbf{x}_1 d\mathbf{x}_2 \cdots d\mathbf{x}_N \hat{P}_n \{\chi_i^*(\mathbf{x}_1)\chi_j^*(\mathbf{x}_2)\cdots\} \hat{v}(\mathbf{r}_1, \mathbf{r}_2) (1 - \hat{P}_{12}) \hat{P}_n \{\chi_k(\mathbf{x}_1)\chi_j(\mathbf{x}_2)\cdots\}. \end{aligned}$$

Since χ_i is orthogonal to any other spin orbital, it will have to be occupied by either electron 1 or electron 2 for a term to be non-zero. If χ_1 is occupied with electron 1, then electron 2 can occupy any of the remaining $N - 1$ spin orbitals common to $|K\rangle$ and $|J\rangle$, and visa versa. The remaining $N - 2$ electrons can be arranged $(N - 2)!$ different ways. We get

$$\begin{aligned} \langle K|\hat{v}(\mathbf{r}_1, \mathbf{r}_2)|J\rangle &= \frac{1}{N(N - 1)} \sum_{j \neq i}^N \\ &(\int d\mathbf{x}_1 d\mathbf{x}_2 \chi_i^*(\mathbf{x}_1)\chi_j^*(\mathbf{x}_2) \hat{v}(\mathbf{r}_1, \mathbf{r}_2) (1 - \hat{P}_{12}) \chi_k(\mathbf{x}_1)\chi_j(\mathbf{x}_2) \\ &+ \int d\mathbf{x}_1 d\mathbf{x}_2 \chi_i^*(\mathbf{x}_2)\chi_j^*(\mathbf{x}_1) \hat{v}(\mathbf{r}_1, \mathbf{r}_2) (1 - \hat{P}_{12}) \chi_k(\mathbf{x}_2)\chi_j(\mathbf{x}_1)). \end{aligned}$$

Since both $\hat{v}(\mathbf{r}_1, \mathbf{r}_2) = \hat{v}(\mathbf{r}_2, \mathbf{r}_1)$ and $\hat{P}_{12} = \hat{P}_{21}$, we can exchange the integration variables in the last term. We get

$$\begin{aligned} \langle K | \hat{v}(\mathbf{r}_1, \mathbf{r}_2) | J \rangle &= \frac{2}{N(N-1)} \sum_{j \neq i}^N \\ &\cdot \int d\mathbf{x}_1 d\mathbf{x}_2 \chi_i^*(\mathbf{x}_1) \chi_j^*(\mathbf{x}_2) \hat{v}(\mathbf{r}_1, \mathbf{r}_2) (1 - \hat{P}_{12}) \chi_k(\mathbf{x}_1) \chi_l(\mathbf{x}_2). \end{aligned}$$

The matrix element of the full two-electron operator in Eq. 23 is thus

$$\langle ij \cdots | \hat{v}(\mathbf{r}_1, \mathbf{r}_2) | kl \cdots \rangle = \sum_j^N [\langle ij | \hat{v} | kj \rangle - \langle ij | \hat{v} | jl \rangle],$$

where the restriction on the sum is removed since $[\langle ii | \hat{v} | ki \rangle - \langle ii | \hat{v} | ik \rangle] = 0$.

For the final scenario we can follow the same argument as before and write the matrix element as

$$\begin{aligned} \langle ij \cdots | \hat{v}(\mathbf{r}_1, \mathbf{r}_2) | kl \cdots \rangle &= \frac{1}{N!} \sum_n^{N!} \\ &\cdot \int d\mathbf{x}_1 d\mathbf{x}_2 \cdots d\mathbf{x}_N \hat{P}_n \{ \chi_i^*(\mathbf{x}_1) \chi_j^*(\mathbf{x}_2) \cdots \} \hat{v}(\mathbf{r}_1, \mathbf{r}_2) (1 - \hat{P}_{12}) \hat{P}_n \{ \chi_k(\mathbf{x}_1) \chi_l(\mathbf{x}_2) \cdots \}. \end{aligned}$$

Since χ_i and χ_j is orthogonal to any other spin orbital, they will have to be occupied by electron 1 and electron 2. Further, there are $(N-2)!$ ways to arrange the remaining $(N-2)$ electrons. This gives

$$\begin{aligned} \langle K | \hat{v}(\mathbf{r}_1, \mathbf{r}_2) | J \rangle &= \frac{1}{N(N-1)} \\ &\left(\int d\mathbf{x}_1 d\mathbf{x}_2 \chi_i^*(\mathbf{x}_1) \chi_j^*(\mathbf{x}_2) \hat{v}(\mathbf{r}_1, \mathbf{r}_2) (1 - \hat{P}_{12}) \chi_k(\mathbf{x}_1) \chi_l(\mathbf{x}_2) \right. \\ &\left. + \int d\mathbf{x}_1 d\mathbf{x}_2 \chi_i^*(\mathbf{x}_2) \chi_j^*(\mathbf{x}_1) \hat{v}(\mathbf{r}_1, \mathbf{r}_2) (1 - \hat{P}_{12}) \chi_k(\mathbf{x}_2) \chi_l(\mathbf{x}_1) \right). \end{aligned}$$

Just as in the previous scenario, we can exchange the variables of the integrations and introduce a factor of $N(N-1)/2$ to get the matrix element of the full two-electron operator, that is

$$\langle ij \cdots | \hat{v}(\mathbf{r}_1, \mathbf{r}_2) | kl \cdots \rangle = \langle ij | \hat{v} | kl \rangle - \langle ij | \hat{v} | lk \rangle.$$

For the same reason as for why Slater determinants with more than one different spin orbitals yielded a zero for the one-electron operator, we get zero for matrix elements where the Slater determinants differ by more than two spin orbitals for the two-electron operators.

C Second quantization anticommutation relations

In this section, we aim to deduce the anticommutation relations for the creation and annihilation operators, shown in Eqs. 4, 5 and 6. We will start with the commutation relation between two creation operators in Eq. 4. We start by applying $a_i^\dagger a_j^\dagger$ and $a_j^\dagger a_i^\dagger$ to the Slater determinant $|kl \cdots \rangle$, that is

$$\begin{aligned} a_i^\dagger a_j^\dagger |kl \cdots \rangle &= |ijkl \cdots \rangle \\ a_j^\dagger a_i^\dagger |kl \cdots \rangle &= |jikl \cdots \rangle = -|ijkl \cdots \rangle, \end{aligned}$$

where we used the antisymmetry of Slater determinants in the final step. We thus have

$$(a_i^\dagger a_j^\dagger + a_j^\dagger a_i^\dagger) |kl \cdots \rangle = |ijkl \cdots \rangle - |jikl \cdots \rangle = 0,$$

which leads to the commutation relation in Eq. 4.

The anticommutation between two annihilation operators in Eq. 5 can be found in a similar fashion. We have

$$\begin{aligned} a_j a_i |ijk \cdots \rangle &= |k \cdots \rangle \\ a_i a_j |ijk \cdots \rangle &= a_i a_j (-|jik \cdots \rangle) = -|k \cdots \rangle, \end{aligned}$$

and the commutation relation in Eq. 5 follows.

For the anticommutation between one creation and one annihilation operator in Eq. 6, we first consider the action of $a_i a_i^\dagger + a_i^\dagger a_i$ on a Slater determinant not occupied with χ_i . We have

$$\begin{aligned} (a_i a_i^\dagger + a_i^\dagger a_i) |\chi_k \cdots \chi_l\rangle &= a_i a_i^\dagger |\chi_k \cdots \chi_l\rangle \\ &= |\chi_k \cdots \chi_l\rangle. \end{aligned}$$

On the other hand, if the Slater determinant is occupied with χ_i , we have

$$\begin{aligned} (a_i a_i^\dagger + a_i^\dagger a_i) |\chi_k \cdots \chi_i \cdots \chi_l\rangle &= a_i^\dagger a_i |\chi_k \cdots \chi_i \cdots \chi_l\rangle \\ &= |\chi_k \cdots \chi_i \cdots \chi_l\rangle. \end{aligned}$$

We thus have

$$\{a_i, a_i^\dagger\} = 1.$$

We finally consider $a_i a_j^\dagger + a_j^\dagger a_i$ when $i \neq j$. The operator can only give a non-zero when the Slater determinant is occupied with χ_i and not with χ_j . We then have

$$\begin{aligned} (a_i a_j^\dagger + a_j^\dagger a_i) |\chi_k \cdots \chi_i \cdots \chi_l\rangle &= -(a_i a_j^\dagger + a_j^\dagger a_i) |\chi_i \cdots \chi_k \cdots \chi_l\rangle \\ &= |\chi_j \cdots \chi_k \cdots \chi_l\rangle - |\chi_j \cdots \chi_k \cdots \chi_l\rangle = 0. \end{aligned}$$

Combining our results gives the anticommutation relation in Eq. 6, that is

$$\{a_i, a_j^\dagger\} = \delta_{ij}.$$

D Functional derivatives

This section gives a brief introduction to functional derivatives. A more comprehensive read can be found in [7]. To obtain the MCTDHF orbital equation, we have to explain how to perform functional derivatives. A functional is a rule which associates a number with a set of functions, $f_1, f_2, \cdots, f_n \rightarrow F[f_1, f_2, \cdots, f_n]$. One form of a functional is

$$F[f] = \int \omega(x) f(x) dx, \quad (24)$$

where $\omega(x)$ is some weight function. We want to find the rate of change of the above functional in the direction an arbitrary function $\phi(\mathbf{r})$.

We define the functional derivative through the following equation,

$$\int d\mathbf{r} \frac{\delta F[f]}{\delta f(\mathbf{r})} \phi(\mathbf{r}) = \lim_{\epsilon \rightarrow 0} \frac{F[f(\mathbf{r}) + \epsilon \phi(\mathbf{r})] - F[f(\mathbf{r})]}{\epsilon}, \quad (25)$$

where the functional derivative is defined as

$$\frac{\delta F[f]}{\delta f(\mathbf{r})}.$$

We can find the functional derivative of Eq. 24 by inserting its expression in Eq. 25, that is

$$\int d\mathbf{r} \omega(\mathbf{r}) \phi(\mathbf{r}) = \lim_{\epsilon \rightarrow 0} \int d\mathbf{r} \frac{\omega(\mathbf{r}) [f(\mathbf{r}) + \epsilon \phi(\mathbf{r}) - f(\mathbf{r})]}{\epsilon}. \quad (26)$$

Now compare the first expression in Eq. 26 with the first expression in Eq. 25. Since $\phi(\mathbf{r})$ is an arbitrary function, we must have

$$\frac{\delta F[f]}{\delta f(\mathbf{r})} = \omega(\mathbf{r}). \quad (27)$$

One can show that we have the following product rule for functional derivatives

$$\frac{\delta FG[f]}{\delta f(\mathbf{r})} = \frac{\delta F[f]}{\delta f(\mathbf{r})} G[f] + \frac{\delta G[f]}{\delta f(\mathbf{r})} F[f]. \quad (28)$$